HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Assisted in root cause analysis for process deviations by examining scientific data, contributing to the resolution of 3 key technical issues impacting batch yields.
- Participated in cross-functional teams to investigate process issues, documenting findings and contributing to knowledge-based reports summarizing study conclusions.
- Produced and maintained accurate laboratory records in accordance with site procedures, ensuring traceability for over 150 distinct experimental runs.
- Developed Python scripts to automate data extraction and preliminary analysis from laboratory instruments, reducing manual processing time by approximately 40% for research team.
- Applied statistical methods and data visualization techniques to analyze experimental results, supporting the identification of process improvement opportunities.
- Collaborated with senior scientists to support the delivery of process solutions and contribute to laboratory and process validation report packages.
- Containerized data processing scripts using Docker for consistent execution across development and testing environments, ensuring reproducible analytical workflows.

SKILLS

Laboratory Sciences – Process Sciences, Laboratory Investigations, Process Validation, Root Cause Analysis, Deviation Management, Scientific Data Analysis

Programming & Scripting – Python, Bash Scripting, Docker

Data Analysis & Visualization – R, RStudio, pandas, NumPy, Power BI, Tableau, DAX, Power Query, Statistical Analysis, Exploratory Data Analysis

Databases & Data Warehousing – SQL, PostgreSQL, MySQL, Oracle, Data Modelling, Data Warehousing

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Jenkins, Git, CI/CD

Enterprise Systems – Jira, Confluence, ERP Systems Support, Change Management

PROJECTS

Automated Data Pipeline for Bio-Process Monitoring

- Engineered a Python-based data pipeline to ingest and process experimental data from laboratory instruments, handling approximately 5,000 data points per experiment.
- Developed automated reporting scripts using pandas and Power BI to visualize key process parameters and experimental outcomes for research scientists.
- Implemented data validation checks to ensure data integrity and consistency, reducing downstream analysis errors by an estimated 15%.

MSc Applied Machine Learning for Predictive Analytics

- Developed predictive models using scikit-learn on a dataset of 150,000 customer behaviour records to forecast engagement likelihood.
- Performed feature engineering and selection to improve model performance, achieving a **30% uplift in prediction accuracy** over baseline models.
- Evaluated model performance using cross-validation and AUC metrics, presenting findings and model insights to faculty and peers.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)